

2012-2013 – Toulouse School of Economics – Master 2

Macro I Syllabus

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The purpose of this course is

- to present the main theories of macroeconomics in the long run, i.e. the theory of economic growth and unemployment,
- to introduce the main facts of the business cycles, and the skeleton of dynamic stochastic general equilibrium models.

In our way, we'll introduce/review a number of tools in dynamic optimization, equilibrium derivation, and welfare analysis.

Prerequisites are : Kuhn-Tucker optimization theory ; standard calculus and linear algebra ; standard micro price theory.

Useful textbooks are:

- Acemoglu, Introduction to modern economic growth, MIT press
- Ljungqvist and Sargent, Recursive Macroeconomic Theory , MIT Press
- Pissarides, Equilibrium Unemployment Theory

And also

- Blanchard and Fischer, Lectures on macroeconomics, MIT Press
- Romer, Advanced Macroeconomics, McGraw Hill
- Aghion and Howitt, The Economics of Growth, MIT Press
- Layard-Nickell-Jackman, Unemployment, Oxford

The tentative plan of the course is:

1. The Ramsey growth model

- Tool : optimal control
- Readings :
 - Blanchard/Fischer, chapter 2
 - Acemoglu, chapter 7
 - Acemoglu, chapter 8
 - Ramsey (1928), " A mathematical theory of saving ", Economic Journal

2. Innovation and endogenous growth

- Tool : the Dixit-Stiglitz aggregative model
- Readings :
 - Acemoglu, chapters 12 and 13
 - Dixit-Stiglitz, " Monopolistic Competition and Optimum Product Diversity" The American Economic Review, Vol. 67, No. 3 (Jun., 1977), pp. 297-308
 - Romer, Paul, ".Endogenous Technological Change ", Journal of Political Economy, Vol. 98, No. 5 (Oct., 1990), pp. S71-S102
 - Grossman-Helpman, " Quality Ladders in the Theory of Growth" Review of Economic Studies, Vol. 58, No. 1 (Jan., 1991), pp. 43-61
 - Aghion-Howitt, " A Model of Growth Through Creative Destruction" Econometrica, Vol. 60, No. 2 (Mar., 1992), pp. 323-351

3. Empirical growth

- Tools : Log-linearization around steady state; computing a speed of convergence
- Readings :
 - Acemoglu, chapter 3, 4
 - Mankiw, Romer, and Weil (1992), " A contribution to the empirics of economic growth ", Quarterly Journal of Economics
 - Barro and Sala-i-Martin (1991), " Convergence ", Journal of Political Economy
 - Jones (1995) " Time Series Tests of Endogenous Growth Models" Quarterly Journal of Economics, Vol. 110, No. 2 (May, 1995), pp. 495-525
 - Acemoglu, Johnson and Robinson (2001), " The Colonial Origins of Comparative Development: An Empirical Investigation" The American Economic Review, Vol. 91, No. 5 (Dec., 2001), pp. 1369-1401

4. Unemployment I : Insider-Outsiders and efficiency wages

- Tools: Dynamic programming, discrete and continuous time; 2-state Markov models ; Poisson processes; Asset-pricing interpretation of Bellman equations
- Readings :
 - Lindbeck and Snower, Insiders versus Outsiders The Journal of Economic Perspectives, Vol. 15, No. 1 (Winter, 2001), pp. 165-188

- Grout (1984), " Investment and Wages in the Absence of Binding Contracts: A Nash Bargaining Approach" *Econometrica*, Vol. 52, No. 2 (Mar., 1984), pp. 449-460
- Shapiro and Stiglitz (1984) " Equilibrium unemployment as a worker's discipline device ", *American Economic Review*
- Blanchard and Summers (1986), " Hysteresis and the European Unemployment problem ", *NBER Macroeconomics Annual*
- Layard and Nickell (1990), *Unemployment*, Oxford U. Press
- Ljungqvist and Sargent, chapters 3, 4, 5, 6

5. Unemployment II : The Mortensen/Pissarides matching model

- Tool : Comparing optimum and equilibrium in dynamic models with market failures
- Readings :
 - Pissarides, C. (1990) *Equilibrium Unemployment Theory*, 2nd edition, MIT Press, 1998
 - Mortensen, D. and C. Pissarides, (1994) " Job Creation and Job Destruction in the Theory of Unemployment", *Review of Economic Studies*; 61(3), pages 397-415.
 - Hosios, Arthur (1990), " On the Efficiency of Matching and Related Models of Search and Unemployment" *Review of Economic Studies*, Vol. 57, No. 2 (Apr., 1990), pp. 279-298
 - Moen, Ragnar (1995), " Competitive search equilibrium ", *Journal of Political Economy*

6. Empirical Unemployment

- Readings :
 - Nickell, Stephen. 'Labour Market Institutions and Unemployment in OECD Countries.' *CESifo DICE Report* 1, no. 2 (2003), pp. 13-26
 - Blanchard, Olivier, "the medium run", *Brookings Papers on Economic Activity*, Vol. 1997, No. 2 (1997), pp. 89-158
 - O. Blanchard and J. Wolfers. (2000), "The Role of Shocks and Institutions in the Rise of European Unemployment: the Aggregate Evidence". *Economic Journal*, (462), 2000.

7. Overlapping generations

- Tool : Comparing optimum and equilibrium in dynamic models with market failures
- Readings :
 - Acemoglu, Chapter 9
 - Ljungqvist and Sargent, Chapter 9
 - Allais, Maurice (1947), *Economie et Intérêt*
 - Blanchard, Olivier (1985), "Debts, Deficits, and Finite Horizons", *Journal of Political Economy*, 93:223-247
 - Diamond, Peter (1965), "National debt in a neo-classical growth model", *American Economic Review*, 55: 1126-1150
 - Malinvaud, Edmond (1987), "The overlapping generations model in 1947", *Journal of Economic Literature*

- Samuelson, Paul (1958), "An exact consumption-loan model of interest with or without the social contrivance of money", *Journal of Political Economy* 66: 467-482

8. Business Cycle facts

- Tool : Spectral Analysis, National Accounting
- Readings :
 - Cochrane, John, "Time Series for Macroeconomics and Finance", lectures notes
 - Granger, C, "The Typical Spectral Shape of an Economic Variable", *Econometrica*, 1966
 - Baxter, Marianne and Robert G. King. "Measuring Business Cycles: Approximate Band-Pass Filters For Economic Time Series," *Review of Economics and Statistics*, 1999,
 - Christiano and Fitzgerald, 2003. "The Band Pass Filter," *International Economic Review*
 - Canova, Fabio, 1998. "Detrending and business cycle facts," *Journal of Monetary Economics*
 - Burnside, 1998. "Detrending and business cycle facts: A Comment," *Journal of Monetary Economics*
 - Ravn, Morten O. and Uhlig, Harald, 2001. "On Adjusting the HP-Filter for the Frequency of Observations," *CEPR Discussion Papers* 2858,
 - NIPA Handbook: Concepts and Methods of the U.S. National Income and Product Accounts, 2011, Bureau of Economic Analysis website

9. Real Business Cycles

- Tool : Linearization, General Equilibrium
- Readings :
 - Kydland, Finn E and Prescott, Edward C, 1982. "Time to Build and Aggregate Fluctuations," *Econometrica*
 - King, Robert G. and Plosser, Charles I. and Rebelo, Sergio T., 1988. "Production, growth and business cycles : I. The basic neoclassical model," *Journal of Monetary Economics*,
 - King, Robert G. and Plosser, Charles I. and Rebelo, Sergio T., 1988. "Production, growth and business cycles : II. New directions," *Journal of Monetary Economics*,
 - King, Robert G. and Rebelo, Sergio T., 1999. "Resuscitating real business cycles," *Handbook of Macroeconomics*, in: J. B. Taylor and M. Woodford (ed.), *Handbook of Macroeconomics*, edition 1, volume 1, chapter 14, pages 927-1007 Elsevier.
 - Rogerson, Richard, 1988. "Indivisible labor, lotteries and equilibrium," *Journal of Monetary Economics*
 - Hansen, Gary D., 1985. "Indivisible labor and the business cycle," *Journal of Monetary Economics*

10. Asset Pricing in General Equilibrium

- Readings :
 - Mehra, Rajnish and Prescott, Edward C., 1985. "The equity premium: A puzzle," *Journal of Monetary Economics*

- Jermann, Urban J., 1998. "Asset pricing in production economies," *Journal of Monetary Economics*,
- Kocherlakota, N., 1996. "The Equity Premium: It's Still a Puzzle," *Journal of Economic Literature*
- Rouwenhorst, G., Asset pricing implications of equilibrium business cycle models, in: T.F. Cooley (ed) *Frontiers of Business Cycle Research* chapter 10, Princeton University Press (1995)
- Alvarez and Jermann, 2004. "Using Asset Prices to Measure the Cost of Business Cycles," *Journal of Political Economy*,

11. Inequalities and Empirical Models of Idiosyncratic Risk

• Readings :

- Jonathan Heathcote, Fabrizio Perri and Giovanni L. Violante, 2010. "Unequal We Stand: An Empirical Analysis of Economic Inequality in the United States: 1967-2006," *Review of Economic Dynamics*,
- Mace, Barbara J, 1991. "Full Insurance in the Presence of Aggregate Uncertainty," *Journal of Political Economy*,
- Cochrane, John H, 1991. "A Simple Test of Consumption Insurance," *Journal of Political Economy*,
- Townsend, Robert M, 1994. "Risk and Insurance in Village India," *Econometrica*,
- Richard Blundell, Luigi Pistaferri and Ian Preston, 2008. "Consumption Inequality and Partial Insurance," *American Economic Review*,
- Attanasio, Orazio and Davis, Steven J, 1996. "Relative Wage Movements and the Distribution of Consumption," *Journal of Political Economy*,
- Susan Dynarski and Jonathan Gruber, 1997. "Can Families Smooth Variable Earnings?," *Brookings Papers on Economic Activity*,